

WEEK 4 – ASSIGNMENT

Azure Cloud Fundamentals and Data Pipeline Implementation using Azure Data Factory

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Internship: Celebal Technologies – Data Engineering Internship 2026

Assignment: Assignment 04 – Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective

The objective of this assignment was to understand the fundamentals of Microsoft Azure Cloud and build a complete end-to-end data pipeline using Azure Storage Account and Azure Data Factory (ADF). The pipeline validates a source CSV file using the Get Metadata activity and copies it from a source Blob container to a destination Blob container. This assignment also covers Azure RBAC (Role-Based Access Control) to securely connect Azure services.

Task 1: Azure Portal Exploration & Resource Group Creation

Objective

Create a Resource Group to organize all Azure resources required for the project.

Steps Performed

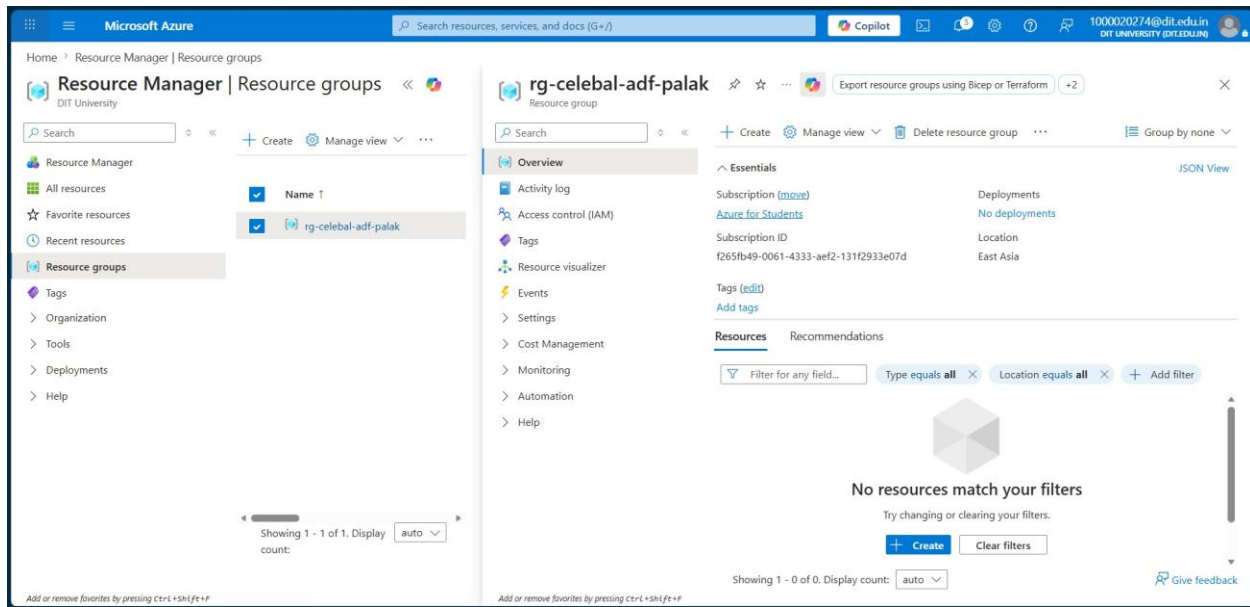
- Logged into the Microsoft Azure Portal using the Azure Student subscription.
- Explored the Azure Portal dashboard.
- Created a new Resource Group named **rg-celebal-adf-palak**.
- Selected the appropriate Azure region supported by the subscription.
- Verified that the Resource Group was created successfully.

Result

The Resource Group was successfully created and used to organize all Azure resources for the assignment.

Screenshot

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Task 2: Azure Storage Setup

Objective

Create Azure Blob Storage and upload the dataset.

Steps Performed

- Created a Storage Account named **stpalakadf2026**.
- Selected Standard Performance and Locally Redundant Storage (LRS).
- Opened the Storage Account after deployment.
- Created a Blob Container named **source**.
- Uploaded the **Sample-Superstore.csv** dataset into the container.

Result

Azure Blob Storage was successfully configured and the dataset was uploaded successfully.

Screenshots

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The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and a user profile icon. The main content area is divided into three sections:

- superstore-data**: A container overview page showing a list of blobs. The 'Sample - Superstore.csv' blob is selected, displaying its properties: URL, LAST MODIFIED (7/11/2026, 9:10:07 PM), CREATION TIME (7/11/2026, 9:10:07 PM), VERSION ID, TYPE (Block blob), SIZE (2.18 MiB), ACCESS TIER (Hot (inferred)), ACCESS TIER LAST MODIFIED (N/A), ARCHIVE STATUS, REHYDRATE PRIORITY, SERVER ENCRYPTED, ETAG (0x8DED62AF2D6994), VERSION-LEVEL IMMUTABILITY POLICY (Disabled), CACHE-CONTROL, CONTENT-TYPE (text/csv), CONTENT-MD5 (swZpBemrTud0U9Rz2/IMA...), CONTENT-ENCODING, and CONTENT-LANGUAGE.
- superstore-data**: A container overview page showing a list of blobs. The 'Sample - Superstore.csv' blob is selected, displaying its properties: URL, LAST MODIFIED (7/11/2026, 9:10:07 PM), CREATION TIME (7/11/2026, 9:10:07 PM), VERSION ID, TYPE (Block blob), SIZE (2.18 MiB), ACCESS TIER (Hot (inferred)), ACCESS TIER LAST MODIFIED (N/A), ARCHIVE STATUS, REHYDRATE PRIORITY, SERVER ENCRYPTED, ETAG (0x8DED62AF2D6994), VERSION-LEVEL IMMUTABILITY POLICY (Disabled), CACHE-CONTROL, CONTENT-TYPE (text/csv), CONTENT-MD5 (swZpBemrTud0U9Rz2/IMA...), CONTENT-ENCODING, and CONTENT-LANGUAGE.
- stpalakadf2026_1783784037291 | Overview**: A deployment overview page showing the deployment status as 'Your deployment is complete'. It includes details such as Deployment name, Subscription, Resource group, Start time, and Correlation ID. The page also features a 'Go to resource' button and a list of next steps.

Task 3: Azure Data Factory Configuration

Objective

Configure Azure Data Factory and establish connectivity with Azure Blob Storage.

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Steps Performed

- Created a new Azure Data Factory instance named **adfpalak2026**.
- Opened the ADF Studio interface.
- Explored the Author, Monitor, and Manage sections.
- Created a Linked Service connecting Azure Data Factory with Azure Blob Storage.
- Created the Source Dataset for the uploaded CSV file.
- Created the Destination Dataset for the output container.

Result

Azure Data Factory was successfully configured and connected to Azure Storage.

Screenshots

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Microsoft Azure | Data Factory | adfpalak2026

Factory Resources

- Pipelines: 0
- Change Data Capture (preview): 0
- Datasets: 1
 - DS_Destination_Superstore
 - DS_Source_Superstore
- Data flows: 0
- Power Query: 0

DelimitedText
DS_Destination_Superstore

Connection Schema Parameters

Linked service * LS_BlobStorage

File path * superstore-data / Directory / Superstore_Output.csv

Compression type No compression

Column delimiter Comma (,)

Row delimiter Default (\r,\n, or \r\n)

Encoding Default(UTF-8)

Quote character Double quote (")

Escape character Backslash (\)

Properties

General Related

Name * DS_Destination_Superstore

Description

Annotations

+ New

Cancel

Microsoft Azure | Data Factory | adfpalak2026

Factory Resources

- Pipelines: 0
- Change Data Capture (preview): 0
- Datasets: 1
 - DS_Source_Superstore
- Data flows: 0
- Power Query: 0

DelimitedText
DS_Source_Superstore

Connection Schema Parameters

Linked service * LS_BlobStorage

File path * superstore-data / Directory / Sample - Superstore.csv

Compression type No compression

Column delimiter Comma (,)

Row delimiter Default (\r,\n, or \r\n)

Encoding Default(UTF-8)

Quote character Double quote (")

Escape character Backslash (\)

Properties

General Related

Name * DS_Source_Superstore

Description

Annotations

+ New

Cancel

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1

1000020274@dit.edu.in
DIT UNIVERSITY

New linked service

Azure Blob Storage [Learn more](#)

Authentication type

Account key

Connection string

Azure Key Vault

Account selection method ⓘ

☒ From Azure subscription ☐ Enter manually

Azure subscription ⓘ

Azure for Students (f265fb49-0061-4333-aef2-131f2933e07d)

Storage account name *

stpalakadf2026

Additional connection properties

+ New

Test connection ⓘ

☒ To linked service ☐ To file path

Annotations

+ New

> Parameters

✓ Connection successful

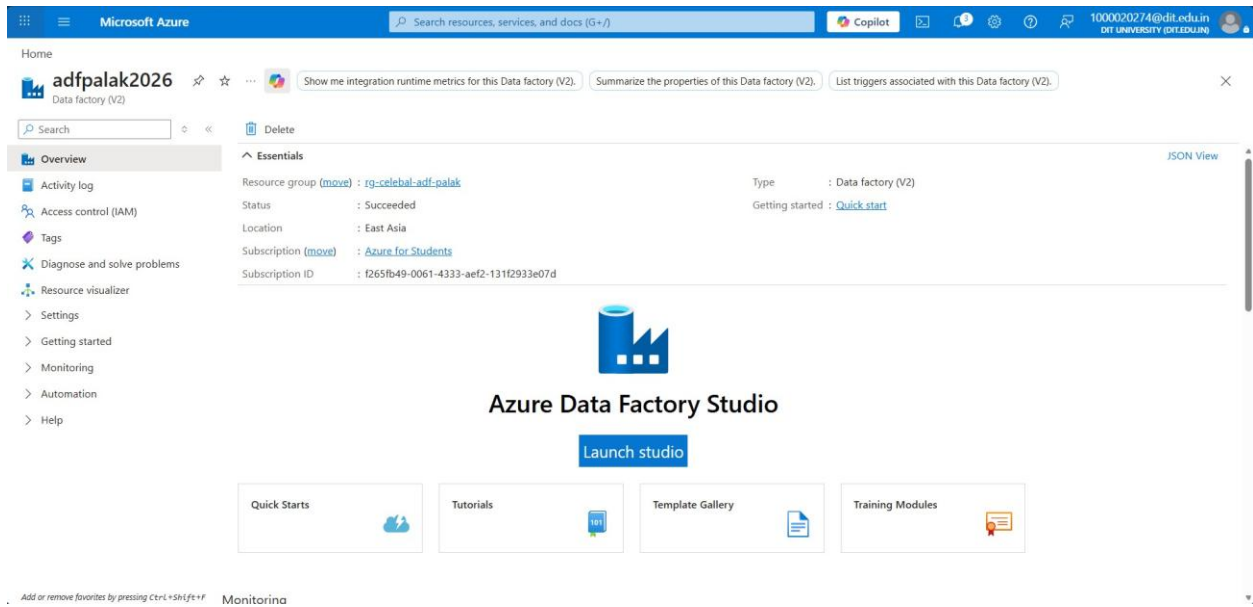
Test connection

Cancel

Create

Back

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Task 4: Pipeline Development

Objective

Build an end-to-end Azure Data Factory pipeline.

Steps Performed

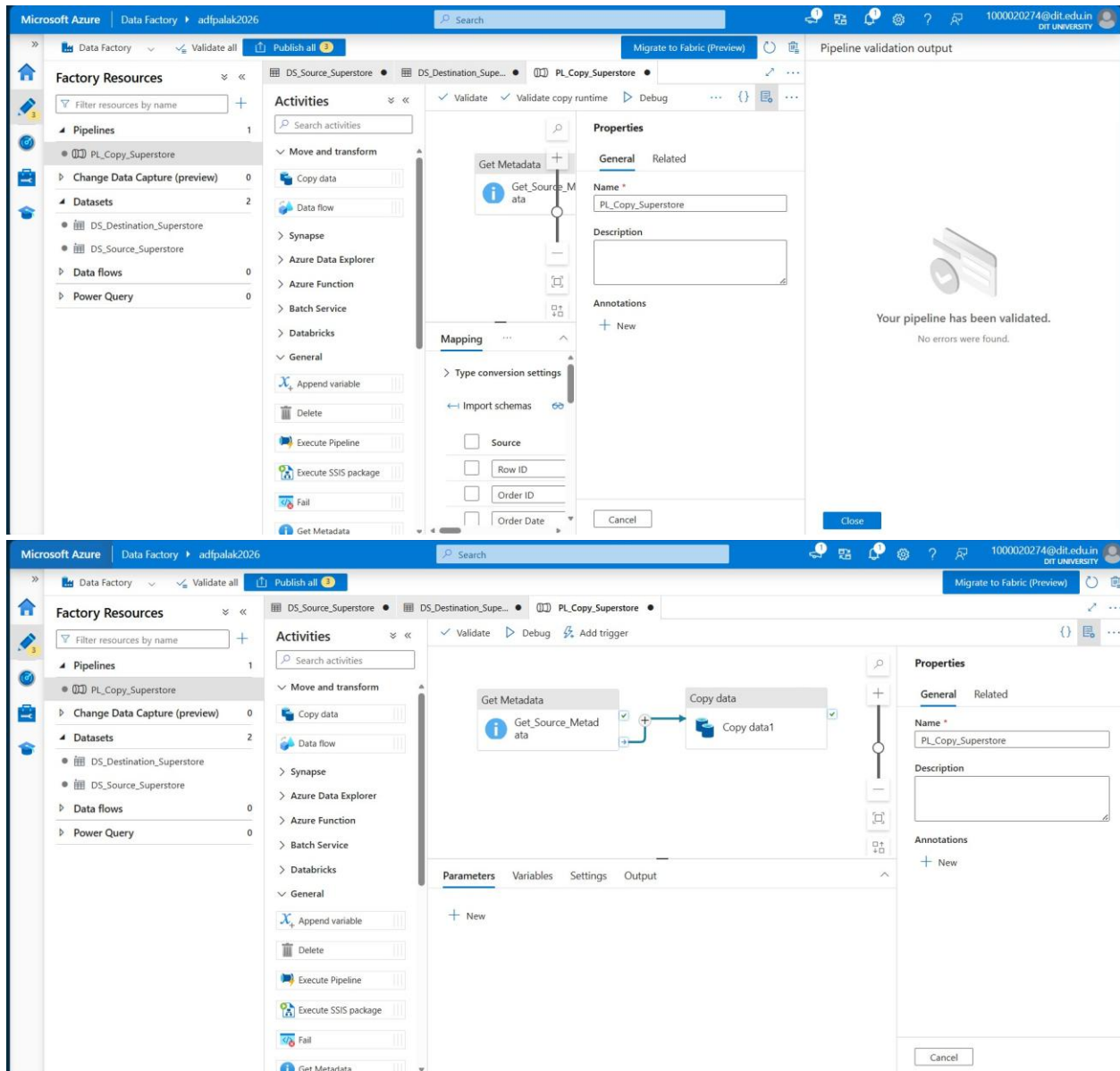
- Created a new pipeline named **PL_Copy_Superstore**.
- Added the **Get Metadata** activity to validate the source CSV file.
- Configured the **Copy Data** activity.
- Connected the Source Dataset with the Destination Dataset.
- Validated the pipeline before execution.

Result

The pipeline was successfully designed and configured.

Screenshots

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Task 5: Pipeline Execution

Objective

Execute the Azure Data Factory pipeline and verify successful execution.

Steps Performed

- Executed the pipeline using **Debug**.
- Published all pipeline changes.
- Triggered the pipeline manually.
- Opened the Monitor section to verify execution.

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- Confirmed that the pipeline completed successfully without any errors.
- Verified that the output CSV file was copied to the destination Blob container.

Result

The pipeline executed successfully and copied the data from the source container to the destination container.

Screenshots

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Pipeline validation output



Your pipeline has been validated.

No errors were found.

Close

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The image displays two screenshots from the Azure portal. The top screenshot shows the Azure Data Factory interface for a pipeline named 'PL_Copy_Superstore'. The pipeline consists of two activities: 'Get Metadata' (Get_Source_Metadata) and 'Copy data' (Copy data1). The pipeline status is 'Succeeded'. The bottom screenshot shows the Azure Storage Explorer interface for a container named 'superstore-data'. It displays a table of blobs, including 'Sample - Superstore.csv'.

Activities

- Copy data
- Data flow
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General
- Append variable
- Delete
- Execute Pipeline
- Execute SSIS package
- Fail
- Get Metadata

Parameters **Variables** **Settings** **Output**

Pipeline run ID: 54f59c4b-0289-4d41-bcab-a996
Pipeline status: Succeeded

Showing 1 - 2 of 2 items

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime
Copy data1	Succeeded	Copy data	7/11/2026, 10:13:36 PM	15s	AutoResolveIntegrationRuntime (Eas
Get_Source_Metadata	Succeeded	Get Metadata	7/11/2026, 10:13:30 PM	5s	AutoResolveIntegrationRuntime (Eas

Microsoft Azure

Home > stpalakad2026 | Containers

superstore-data

Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - Superstore.csv	11/7/2026, 10:18:47 pm	Hot (Inferred)	Block blob	2.18 MiB	Available

adfpalak2026

Search

Pipeline runs

Triggered Debug Rerun Cancel options Refresh Edit columns List Gantt

Filter by run ID or name

Chennai, Kolkata, Mu... : Last 24 hours

Pipeline name: All

Status: All

Runs: Latest runs

Copy filters Export to CSV

Showing 1 - 1 items

Last refreshed 1 minute ago

Pipeline name	Run start	Run end	Duration	Triggered by	Status	Run	Parameters
PL_Copy_Superstore	7/11/2026, 10:18:21 PM	7/11/2026, 10:18:51 PM	30s	Manual trigger	Succeeded	Original	

Task 6: IAM Role Assignment

Objective

Grant Azure Data Factory permission to securely access Azure Blob Storage using Azure Role-Based Access Control (RBAC).

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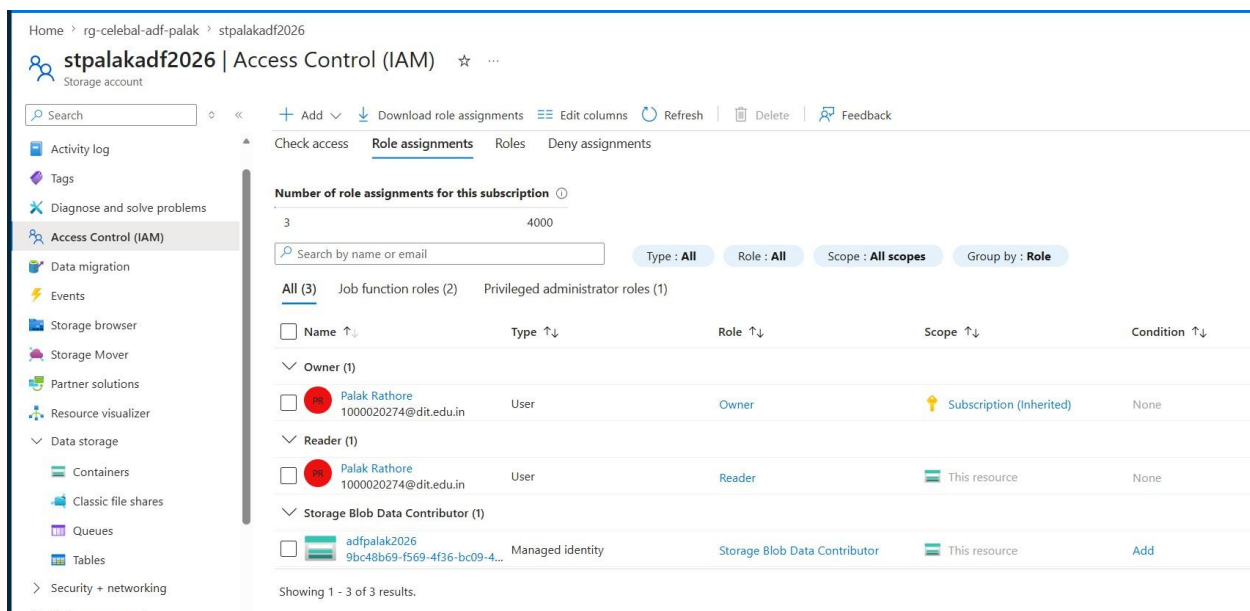
Steps Performed

- Opened the Azure Storage Account.
- Navigated to **Access Control (IAM)**.
- Assigned the **Reader** role to my Azure account.
- Assigned the **Storage Blob Data Contributor** role to the Azure Data Factory Managed Identity (**adfpalak2026**).
- Verified that the required permissions were successfully assigned.

Result

Azure Data Factory was granted permission to access Azure Blob Storage securely. The Managed Identity can now read and write files between Blob containers.

Screenshot



Mini Project: End-to-End Data Pipeline

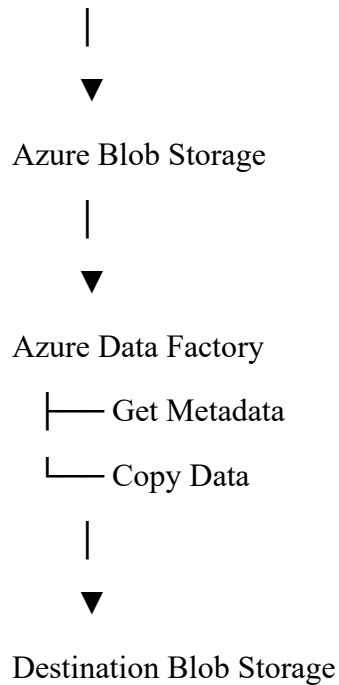
Problem Statement

Build a complete Azure Data Factory pipeline that reads a CSV file from Azure Blob Storage, validates the file using the Get Metadata activity, and copies the file to another Blob Storage location.

Pipeline Architecture

Sample-Superstore.csv

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Workflow

1. Uploaded the Superstore dataset into Azure Blob Storage.
2. Created Linked Services connecting Azure Data Factory to Azure Storage.
3. Created Source and Destination datasets.
4. Added the Get Metadata activity to validate the source file.
5. Configured the Copy Data activity.
6. Executed the pipeline using Debug and Trigger.
7. Monitored pipeline execution through Azure Data Factory Monitor.
8. Verified that the output file was copied successfully.
9. Configured IAM roles for secure access.

Final Output

- Source CSV uploaded successfully.
- Metadata validated successfully.
- Pipeline executed successfully.
- Data copied successfully to the destination Blob container.
- IAM permissions configured successfully.

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- End-to-end Azure Data Factory pipeline implemented successfully.

Learning Outcomes

Through this assignment, I gained practical experience with Microsoft Azure services and Azure Data Factory. I learned how to create and manage Azure resources, configure Azure Storage Accounts, create Blob containers, upload datasets, build Linked Services and Datasets in Azure Data Factory, design data pipelines using Get Metadata and Copy Data activities, execute and monitor pipelines, assign IAM roles using Azure RBAC, and verify successful data movement between Blob Storage containers. This assignment provided a clear understanding of building secure and scalable cloud-based data integration pipelines.

Conclusion

This assignment successfully demonstrated the implementation of an end-to-end Azure Data Factory pipeline. The project covered Azure resource creation, Blob Storage configuration, Azure Data Factory pipeline development, metadata validation, secure access management using IAM roles, and successful data transfer between Blob Storage containers. The objectives of the assignment were achieved successfully, providing hands-on experience with Azure cloud services and modern data engineering workflows.